

Vishwas Kumar

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PROFESSIONAL SUMMARY

Backend-leaning MERN Developer focused on building scalable, production-oriented applications with **TypeScript**, Node.js, and MongoDB. Skilled in designing secure REST APIs, **JWT/RBAC** authentication, and optimized database queries. Experienced with Redis queue-based systems, **Docker**, and **CI/CD** workflows.

TECHNICAL SKILLS

Languages & Backend: JavaScript (ES6+), TypeScript, HTML5, CSS3.

Frontend: React.js, React Query, Tailwind CSS, shadcn/ui, React Hook Form, Zod, Framer Motion.

Backend: Node.js, Express.js, REST APIs, JWT Authentication, RBAC, BullMQ, Redis, Rate Limiting, Helmet.

Database: MongoDB, Mongoose, Aggregation Pipelines, Indexing Strategies.

Testing & DevOps: Vitest, Supertest, MongoMemoryServer, Docker, Github Actions (CI/CD).

Tools & Concepts: Git, GitHub, Postman, Render, Vercel, Scalable Backend Architecture, Queue-Based Systems, API Security, Idempotency, Pagination, Concurrency Handling.

PROJECTS

PulseCheck – Distributed Uptime Monitoring System | [Live](#) | [GitHub](#)

- Built a distributed uptime monitoring system using an asynchronous queue-worker architecture (**API → Queue → Worker → Database**) for scalable heartbeat processing.
- Implemented BullMQ workers with retry strategies, **exponential backoff**, and **idempotent** processing for concurrency-safe operations.
- Reduced dashboard summary latency by **~90%** by replacing runtime aggregation queries with precomputed hourly metrics.
- Optimized cursor-based pagination using MongoDB indexing strategies (covered, partial, ESR), validated with **10k+** seeded records using **IXSCAN** query plans.

Tech: Node.js, Express, MongoDB, Redis, BullMQ, React, TypeScript, Docker

Fitness Planner – Scalable Backend Architecture | [Live](#) | [GitHub](#)

- Built a fitness planner with a layered architecture and Zod validation, using **score-based** exercise filtering and reusable service-layer logic to generate personalized workout plans.
- Implemented JWT authentication, refresh token rotation, **RBAC** (Admin/User), and protected routes for secure multi-role access control.
- Wrote **48** unit and integration tests using Vitest, Supertest, and MongoMemoryServer covering authentication flows, validation, cookies, and error handling.

Tech: React, TypeScript, Node.js, Express, MongoDB, JWT, Zod

FilmBox – Performance-Focused Frontend Application | [Live](#) | [GitHub](#)

- Built a movie discovery and watchlist application with infinite scrolling and optimized data fetching.
- Implemented React Query caching and **prefetching** to eliminate redundant API requests, decreasing client-side network overhead by **40%**.
- Enhanced accessibility using **ARIA** attributes and responsive UI patterns for improved usability.

Tech: React, TypeScript, React Query, Tailwind CSS

EDUCATION

B.Tech – Computer Science & Engineering (AIML)

August 2022 – June 2026

Rungta College of Engineering & Technology | CGPA: 7.09